

Blockchain-Assisted Revocable Cross-Domain Authentication for Vehicular Ad-Hoc Networks

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Abstract—With the rapid development of vehicular ad-hoc networks (VANETs) and the increasing diversification of user demands, interactions between different management domains have become more frequent. Identity authentication is an effective way to establish cross-domain trust and secure communication. However, the existing cross-domain authentication schemes of VANETs are limited to the same management or authentication technology for each domain and rely on centralized cross-domain identity management. Even distributed management solutions encounter latency sensitivity, security and privacy challenges. To address these challenges, we propose a blockchain-assisted revocable cross-domain authentication scheme for VANETs. The proposed scheme can establish trust between domain entities by deploying different authentication methods and using distributed management to avoid single-point failures. In addition, the scheme can revoke the identity of malicious vehicles by updating the group public key, thereby ensuring the security and privacy of cross-domain Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. This design avoids the additional impacts of blockchain technology constraints on the high mobility and real-time requirements of VANETs. Security analysis and performance evaluation show that our scheme can resist more attacks and has better security than other related schemes while also achieving a better balance between communication and computational cost.

Index Terms—VANETs, cross-domain authentication, conditional privacy-preserving, blockchain, smart contract.

I. INTRODUCTION

AS AN indispensable component of the Internet of Things (IoT), vehicular ad-hoc networks (VANETs) play an increasingly vital role in our daily lives [1]. Because entities such as vehicles and infrastructure are typically managed by different domain management centers or authorities, the authentication

technology within each domain is different. A group of devices belonging to the same management domain constitutes an organization known as a “trust domain.” [2] Entities within the same domain are coordinated by an intra-domain management center, which creates pre-established trust relationships with vehicles and infrastructure, enabling mutual authentication and sharing of application resources. However, when a vehicle moves to another trust domain, trust between the two domains is difficult to build, so this relationship becomes ineffective [3]. Consequently, with the rapid evolution of VANETs and diversification of user demands, the need for access to application resources between different trust domains has arisen, including different cities and even countries [4].

In VANETs, the primary communication types are Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Network (V2N), and Vehicle-to-Pedestrian (V2P). In-vehicle on-board units (OBUs) interact with other entities within or foreign to the domain through protocols like IEEE 802.11p or cellular communication networks (such as 4G and 5G) to facilitate transportation and entertainment-related applications [5]. Due to the openness of such wireless communication channels, security and privacy issues have arisen [6], [7], [8]. For example, attackers can monitor, intercept, or tamper with real-time messages in a vehicle communication channel, thereby posing a threat to road traffic safety or driver privacy. Furthermore, there is a lack of trust between different domains. Therefore, when vehicles access cross-domain services, the accessed domain must authenticate the vehicle’s identity. Only after authentication can a vehicle communicate with other entities within the domain to ensure the security of the VANET communication system. However, V2V and V2I communications across different trust domains still face problems in terms of security authentication.

Currently, for the sake of manageability existing VANETs cross-domain authentication solutions are limited to the same or similar types of authentication or management techniques in each domain [9], [10], or it is assumed that management centers trust each other in advance. Many studies on secure communication between cross-domain entities, such as cross-domain vehicles and roadside units, are based on encryption technologies, such as group signatures [11], [12] and anonymous identity authentication [13], [14], [15]. They rely on centralized, fully-trusted authorities to manage cross-domain vehicles [16], [17], which increases an attacker’s determination to launch attacks to compromise the private keys of the system and makes it susceptible to single-point failures.

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To address the shortcomings of centralized cross-domain identity management schemes, recent decentralized VANET cross-domain authentication solutions based on blockchain have been proposed. For example, Yang et al. [18] introduced a multi-domain authentication scheme for vehicular networks that ensures conditional privacy-preserving. They utilized blockchain to facilitate cross-domain information sharing and designed a pseudonym distribution mechanism to guarantee vehicle anonymity and traceability. Li et al. [19] addressed the issue of linkability in existing blockchain-based vehicular network authentication schemes by proposing a collusion-resistant, unlinkable authentication key protocol. This protocol achieves multi-trust authority (TA) domain V2V and V2I communication through different tickets, pseudonyms, and signatures.

However, even with the incorporation of blockchain technology require third-party assistance in VANETs communication, or these solutions face limitations inherent to the technology itself. For instance, communication entities within VANETs must query the blockchain in real-time for mutual identity authentication, leading to increased communication latency that does not meet the latency-sensitive requirements of highly mobile vehicles. Additionally, cross-domain vehicles must ensure their privacy, and therefore, they must communicate anonymously to avoid being traced or attacked by malicious entities. While protecting user privacy, there is also a need for trustworthy authorities or distributed management systems capable of tracing and revoking the identities of wrongdoers to ensure stable operation of the entire vehicular network. Unfortunately, none of the currently available cross-domain authentication solutions for VANETs satisfy these requirements.

To address these limitations, we propose a blockchain-assisted revocable cross-domain authentication solution for VANETs. The scheme can support authentication between trust domains running different identity management technologies without changing existing intra-domain infrastructure deployments, particularly for scenarios that cross borders. Furthermore, to address the security and privacy issues of vehicle cross-domain communication in VANETs, we employ a short random group signature algorithm. This allows vehicles to generate one-time licenses for communication with entities within the accessed domain, ensuring the integrity, anonymity, unlinkability, and traceability of communicate messages. The contributions of the proposed scheme are as follows:

- 1) To achieve decentralized authentication between entities within different trust domains using different management technologies, blockchain technology is used to distribute information about each trust domain and cross-domain vehicles without the need to assume prior mutual trust between domain management centers. The accessed domain does not require additional storage for cross-domain vehicle information. The vehicle only needs to complete one round of interaction for authentication and obtain a cross-domain license when it first needs to cross domains, which reduce the computational overhead.
- 2) To address the issue of conditional privacy-preserving in cross-domain communication for VANETs, a short random group signature method is used to enable vehicles to

self-generate one-time licenses for anonymous V2V and V2I communication. During this process, no third-party trusted authority or decentralized system is required to assist. The accessed domain can trace the real identity of a malicious vehicle and revoke its identity by updating the domain key and smart contract design. The design of this scheme avoids additional overhead caused by real-time on-chain queries.

- 3) A comprehensive security analysis of the proposed scheme was conducted, confirming its security and conditional privacy-preserving capabilities, particularly against collusion attacks. Experimental simulation results show that our scheme balances the computing and communication costs better than other cross-domain authentication solutions for VANETs.

The remainder of this paper is structured as follows: Section II introduces related works in the field. The system model is presented in Section III. Section IV provides a detailed description of the specific VANETs cross-domain scheme. Sections V and VI present a comprehensive security analysis and performance evaluation, respectively. Finally, Section VII concludes the paper.

II. RELATED WORK

There have been a series of research work on authentication design in different VANETs cross-domain scenarios, which are mainly divided into centralized cross-domain authentication schemes and blockchain-based distributed cross-domain authentication schemes.

In the centralized VANETs cross-domain authentication scheme, a separate centralized server manages the identity of cross-domain vehicles. Sun et al. [16] proposed a cross-domain authorized secure data sharing scheme for autonomous vehicle platoons. This scheme ensures secure communication between two different platoons by designing a new ciphertext conversion technology. However, the identity and privacy-preserving of the leader PL is not considered. Lin et al. [17] proposed a fog-based key agreement protocol and authentication scheme among various entities. This framework facilitated cross-domain access to other fog servers. But this scheme exhibited vulnerabilities to impersonation and man-in-the-middle attacks. Vilalta Ricard et al. [20] made significant contributions with a method for vehicle message exchange within cross-border scenarios. The proposed architecture offered valuable insights into the exchange of messages between cross-border vehicles. Although no specific design protocol is given, it provides readers with a good research direction. Similarly, Chen et al. [21] introduced a cloud storage-based scheme for inter-regional vehicle anonymous authentication. This scheme effectively addressed challenges related to the remote server's inability to authenticate vehicle identities. However, it did not offer provisions for tracking vehicle identities. Xu et al. [22] proposed a vehicle cross-domain authentication scheme in LTE-A network based on elliptic curve public key encryption algorithm, which can effectively protect vehicle privacy and security. Yang et al. [23] proposed an edge-assisted decentralized authentication architecture for

VANETs, in which distributed edge nodes authenticate vehicle identities. However, the update process of edge nodes in the scheme is expensive, and there is no identity privacy-preserving for vehicles. Wei et al. [9] proposed an identity authentication and key agreement under the multi-TA model based on fog computing, which solved the problem of system overhead and single-point failure. But this scheme can only resist DoS attack to a certain extent.

In the centralized cross-domain authentication scheme, the centralized trusted authority will be trusted by all cross-domain vehicles. Although easy to design, these schemes always rely on the centralized authority to manage cross-domain identities. Due to authorities are vulnerable to denial of service attacks, which leak all cross-domain user privacy. In response to this issue, some researches have proposed a decentralized cross-domain authentication solution based on blockchain technology.

Lei et al. [3] proposed a key management scheme for Vehicle Communication Systems, which employs blockchain via a secure message network to transmit and verify vehicle keys for cross-border requests. However, this scheme considers the mutual authentication of heterogeneous networks among different Certificate Authorities (CAs). Yang et al. [24] introduced a blockchain-based cross-domain authentication scheme for diverse IoT communication domains' mobile devices within an edge computing environment, achieving distributed management of cross-domain identities. Xue et al. [25] presented a distributed autonomous network cross-domain authentication scheme in a multi-CA setting by utilizing blockchain to offer a cross-domain certificate revocation mechanism. They decouple on-chain and off-chain operations to mitigate the limitations imposed by blockchain technology. Yang et al. [18] proposed a multi-domain authentication solution for vehicular networks that focuses on privacy-preserving. This scheme employs blockchain to enable cross-domain information sharing among multiple management domains. However, it has pseudonym linkability, requiring assistance from Roadside Units (RSUs) and blockchain in V2V scenarios, with RSUs needing to be completely trusted. Chen et al. [26] proposed an anonymous cross-domain authentication scheme called XAuth based on Public Key Infrastructure (PKI). This scheme integrates existing PKI with a certificate transparency system and addresses the limitations of blockchain technology by designing a lightweight verification protocol to enhance response speed. Hua et al. [27] introduced a blockchain-based cross-domain self-authentication scheme to address user identity storage in automated parking systems. However, this scheme involves multiple rounds of interaction between users and servers for authentication. Li et al. [19] presented a collusion-resistant unlinkable VANETs authentication key agreement by using blockchain technology to realize V2V and V2R communication between multiple TA domains through different tickets, different pseudonyms and signatures. Xu et al. [28] designed a blockchain-based RSU-assisted multi-TA cross-domain identity verification and key protocol. All TAs form a multi-server network to collectively manage and store vehicle-related information. However, the scheme requires RSUs to have tamper-resistant devices installed and assumes that TAs do not collude with each other.

In summary, most existing research solutions are unable to achieve cross-domain vehicle authentication between different management domains for VANETs. Seamless switching during cross-domain processes is unattainable, as cross-domain identities are reliant on a single entity for management. Additionally, issues persist with non-anonymous vehicle identities, pseudonym linkability, and the inability to trace and revoke malicious identities. Furthermore, solutions incorporating blockchain technology are constrained by the limitations of the technology itself or require third-party auxiliary communication, which fail to meet the latency-sensitive requirements of VANETs. Therefore, it is necessary to design a decentralized and revocable cross-domain authentication solution that meets the requirement for low-latency communication to ensure the security of VANETs. Blockchain is solely utilized for managing cross-domain identities and does not participate in real-time communication processes, thereby avoiding the impact of the technology's inherent high confirmation latency.

III. PRELIMINARIES AND BACKGROUND

A. Preliminaries

1) *Bilinear Pairing*: Let G_1 , G_2 and G_T be three finite groups of prime order q . g and $\hat{g}$ are the generators of G_1 and G_2 , respectively. Then the map $e : G_1 \times G_2 \rightarrow G_T$ is an asymmetric bilinear map and has the following properties:

- Bilinear: For $a, b \in \mathbb{Z}_p$, there is the relation $e(g^a, \hat{g}^b) = e(g, \hat{g})^{ab}$.
- Non-degenerate: $e(g, \hat{g}) \neq 1_{G_T}$.
- Computable: An efficient method exists to compute the map $e(g^a, \hat{g}^b) = e(g, \hat{g})^{ab}$.

2) *Bivariate Polynomial*: A polynomial of order t in two variables (x, y) is called a bivariate polynomial [29], which satisfies:

$$f(x, y) = \sum_{i,j=0}^t s_{ij} x^i y^j. \quad (1)$$

In formula (1), the coefficient s_{ij} is randomly selected in the finite field $\mathbb{Z}_p^*$, p is a large prime number, and $s_{ij} = s_{ji}$. Among them, the bivariate polynomial has symmetry: $f(x, y) = f(y, x)$. When the cross-domain authentication is successful, a polynomial fragment $g_i(y) = f(id_i, y)$ will be assigned to the vehicle nodes in the network.

3) *Mathematical Assumptions*: *Definition 1*: Computational Diffie-Hellman (CDH) Assumption [30]. Let G_1 be a cyclic group of prime order p and g be a generator of G_1 . For any unknown a', b' , given $g \in G_1$, where $a', b' \in \mathbb{Z}_p^*$, the advantage of any PPT adversary to calculate $g^{a'b'} \in G_1$ without the knowledge of a' and b' is negligible.

Definition 2: LRSW Assumption [31]. Let $X = \hat{g}^{x'}$ and $Y = \hat{g}^{y'}$ in G_2 , where x' and y' are random elements in $\mathbb{Z}_p^*$, let the oracle $\mathcal{O}(m)$ on input $m \in \mathbb{Z}_p$ that chooses a random $h \in G_1$ and outputs a signature $T = (h, h^{x'+my'})$. Then, having access to such oracle and knowing all values but x and y , no adversary can efficiently generate such a valid triple (A, B) such that $A \in G_1 \wedge B = A^{x'+m^*y'}$ and a new m^* not asked to $\mathcal{O}$.

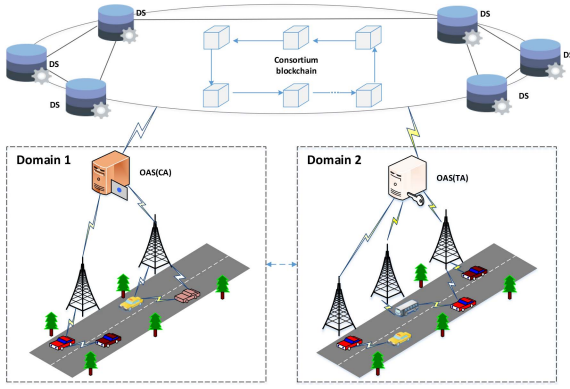


Fig. 1. The system model of our proposed scheme.

B. System Model

Entities involved in the proposed scheme are the original domain authentication server (OAS), roadside unit (RSU), vehicle and dedicated server (DS), as shown in Fig. 1. The entities and their responsibilities are described as follows.

- **OAS:** Original domain authentication server (OAS) refers to the role responsible for managing the identities of vehicles, infrastructure, and other entities within different management domains. For example, it could be a CA managing device certificates in a certificate-based authentication scheme, a Key Generation Center (KGC) in an identity-based authentication scheme, or a TA in other authentication schemes. OAS is considered trustworthy for all entities within its respective management domain. However, there could be a risk of collusion with other OAS entities.
- **RSU:** RSUs within each management domain form a consortium blockchain, providing vehicles with an API to invoke smart contract functions and assisting vehicles to updating relevant licenses. It is assumed that these RSUs will not provide erroneous APIs.
- **Vehicle:** Vehicles are equipped with an OBU and a Tamper-Proof Device (TPD) [26]. OBUs facilitate V2V and V2I communication with other vehicles or roadside infrastructure. TPDs are used to securely store materials and compute secret values. In VANETs, vehicles are considered untrusted entities, hence messages originating from vehicles require identity verification.
- **DS:** Each management domain's OAS selects a DS for cross-domain authentication. All DSs form a consortium blockchain, which offer an API for participating domain's OAS to invoke smart contract functions and aid in the dynamic management of cross-domain vehicle information.

C. Security Requirements

The main security requirements of our proposed protocol are as follows:

- **Mutual authentication:** Our protocol should ensure that the receiver and sender of the message authenticate each other to ensure the authenticity and validity of the message.

TABLE I
NOTATIONS

Notations	Definitions
$G_1, G_2, G_T, g, \hat{g}$	Finite groups G_1, G_2, G_T with order q generated by g and $\hat{g}$
OAS_X^j	The OAS in domain j , X means CA, TA, KGC
V_i^j	i -th vehicle in domain j
GID_j	Domain identifier of OAS
$APK_j / ask_j, U_j / u_j$	Master key of OAS_j^j
$X_j / x_j, Y_j / y_j$	Group key of OAS_j^j
CID_i^j	The cross-domain identity of V_i^j
vpk_i^j	The cross-domain communication public key of V_i^j
ID_i^j	The real identity of V_i^j
Lic_i^j	The cross-domain licence of V_i^j
$E_k(\cdot) / D_k(\cdot)$	Symmetric encryption/decryption using key k
H_1, H_2, H_3	Three secure one-way hash function
ΔT	Validity period of message

- **Conditional privacy-preserving:** Our protocol in the realization of cross-domain vehicle identity anonymity and privacy protection at the same time, to ensure that different two messages of the same vehicle will not be linked on, and TA can be traced back to the malicious vehicle.
- **Resistant to man-in-the-middle attack:** No attacker can intercept and modify the message to deceive the receiver.
- **Resistant to DDoS attack:** Distributed denial-of-service attacker can not attack our protocol.
- **Resistant to collusion attack:** Our protocol should ensure that the access domain OAS, DS server can not complete the collusion attack.
- **Resistant to replay attacks:** Our protocol should ensure that the message is received before checking its freshness.
- **Resistant to impersonation attack:** No attacker can capture and successfully forge the messages sent by the OAS and the vehicle.

D. Threat Model

In the proposed scheme, vehicles communicate with entities in multiple management domains through wireless channels, which exposes them to various common attacks in VANETs. According to the famous Dolev-Yao threat model [32], attacks launched by attackers can be classified into passive and active attacks, including eavesdropping, tampering, impersonation, replay, and man-in-the-middle attacks. In addition, there is a risk of server collusion in multi-domain scenarios. Server collusion refers to the collusion between servers in multiple administrative domains to interfere with the normal access and communication of vehicles in another domain or leak their private data. For example, in cross-domain authentication, the authority of the accessed domain may mark the real identity of the cross-domain vehicles, capture their cross-domain access records, and thereby expose the location privacy of the vehicles. Therefore, OASs in each domain do not always trust each other.

IV. OUR PROPOSED SCHEME

In this section, we describe a cross-domain vehicle authentication scheme designed for VANETs. Table I lists all symbols and their explanations used in the proposed scheme. The first stage

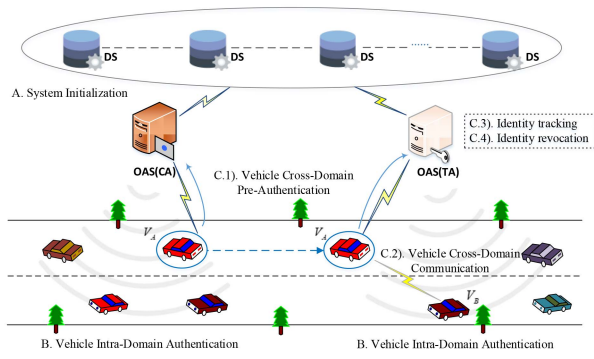


Fig. 2. Framework of our proposed scheme.

is system initialization, where OASs within each management domain negotiate certain common parameters. The second stage is intra-domain authentication, and the scheme can retain the scheme of entity authentication in the original domain, which we call the home domain. The third stage is for vehicle cross-domain authentication, we refer to other domains accessed by vehicles as foreign domains. This phase contains four parts, first is the vehicle cross-domain pre-authentication phase, where cross-domain vehicles interact with OAS in one round with the help of blockchain to obtain cross-domain credentials. Then comes the vehicle cross-domain communication, where the cross-domain vehicle uses the obtained credentials, and randomizes the credentials to communicate with other entities. Then comes vehicle tracking, where the accessed domain OAS can join with the OAS of the vehicle's home domain to track the vehicle's real identity after discovering the malicious behavior. Finally, there is vehicle revocation and group key update, where the accessed domain realizes the complete revocation of the malicious vehicle by updating the group public key in the domain, and thus the legitimate vehicle autonomously updates the cross-domain credentials. The framework flowchart of our proposed scheme is shown in Fig. 2.

A. System Initialization

Before system construction, each management domain must pre-negotiate several common parameters, which are maintained by the OAS of each domain. Furthermore, each domain's OAS possesses a unique domain identifier, GID_j , and needs to generate various public-private key pairs for multi-domain scenarios. For instance, $APK_j = g^{ask_j}$, $U_j = g^{u_j}$ (master key for cross-domain vehicle authentication and malicious vehicle tracing), $X_j = \hat{g}^{x_j}$, $Y_j = \hat{g}^{y_j}$ (group key for when OAS acts as a group administrator during cross-domain vehicle access with entities within the domain as group members). The domain identifier and public key are uploaded to the consortium blockchain formed by DSs. OAS does not require knowledge of the authentication schemes and encryption algorithms used in foreign domains. Additionally, RSUs within each domain maintain a consortium blockchain ledger that records a illegal list of cross-domain vehicles, documenting the real identity IDs of vehicles from

foreign domains that are prohibited from accessing the home domain, denoted as $Illegal_list = \{ID_1, ID_2, \dots, ID_i\}$.

B. Vehicle Intra-Domain Authentication

In this phase, the proposed scheme does not require entities managed within the home domain (such as vehicles or RSUs) to modify their existing authentication schemes. This means that entities within the home domain can continue using their original authentication methods for communication. In contrast to approaches like [18], [26], [19] and [25], which necessitate a unified re-registration and authentication process for vehicles in a multi-domain scenario, our scheme significantly reduces the time cost associated with implementing a unified authentication technology.

C. Vehicle Cross-Domain Authentication

The proposed scheme provides identity verification and communication between any two vehicles, or between a vehicle and an RSU, within two different management domains without altering the authentication mode of the home domain, while ensuring anonymity, unlinkability and traceability of the communication message, and revocability of vehicle identity. It comprises two main components: vehicle cross-domain pre-authentication and vehicle cross-domain communication. For example, the CA deployed in domain A is a certificate-based authentication scheme, and the TA deployed in domain B is an identity-based authentication scheme. Before vehicles in domain A need to cross-domain to domain B, registration and identity verification have been completed in this domain A.

1) *Vehicle Cross-Domain Pre-Authentication*: As the travel routes of vehicles can be predicted in advance, when a vehicle needs to cross multiple domains simultaneously, such as when a vehicle in domain A needs to access domains B, C, D, assuming that the domains managed by OAS utilize ECC signatures and AES encryption algorithms, and vehicle V_i^A has registered its identity and communication public key pk_v with the OAS_{CA}^A of domain A, with the corresponding communication private key sk_v . Additionally, the system private key sk_s has issued a certificate to vehicle V_i^A , and the corresponding system public key is P_{pub} . The cross-domain vehicle authentication phase is illustrated in Fig. 3, where cross-domain vehicle identities are managed by the consortium blockchain composed of DSs.

1) Vehicle first selects three random secret keys $vs k_i^A \in Z_p^*$, and computes $vpk_i^A = g^{vs k_i^A}$ as the cross-domain communication public key, respectively. It then calculates a symmetric key $symk = sk_v \cdot P_{pub}$ and an encrypted value $E_i = E_{symk}(Num, GID_j, ID_i^A, vpk_i^A)$, generates a request message $M_1 = \{E_i, Sign_{v_A}, TS_1\}$, and sends it through a public channel to the OAS_{CA}^A of domain A. Where Num represents the number of domains that vehicle V_i^A needs to access at one time, GID_j represent the domain identifiers of the domains to be accessed, where $j = B, C, D$, $Sign_{v_A}$ represents the signature of the message by vehicle V_i^A using the home domain private key sk_v , and TS_1 represents a timestamp. It's worth noting that vehicle V_i^A needs to request a identity $ID_i^A \in Z_p^*$

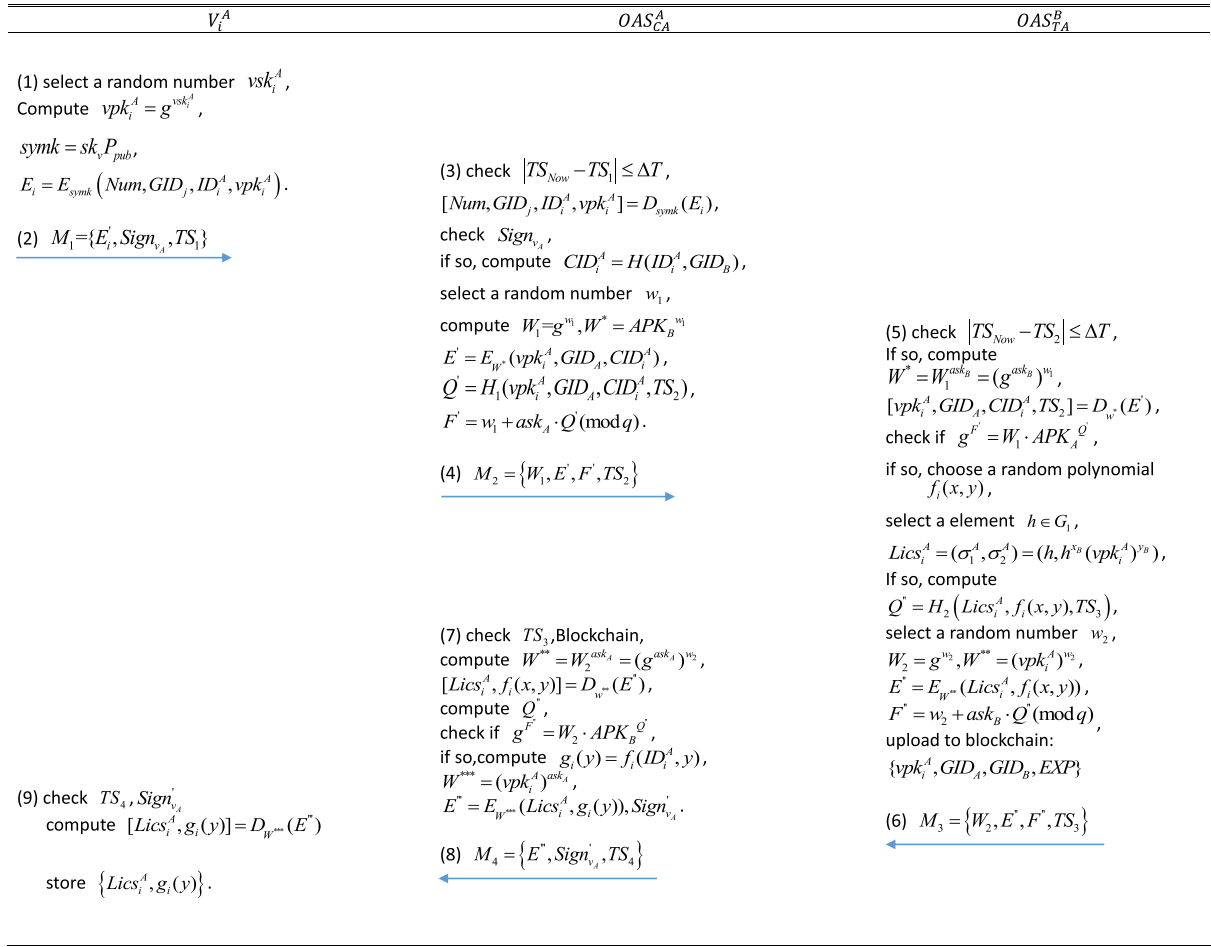


Fig. 3. Cross-domain pre-authentication phase

from OAS_{CA}^A through a secure channel before initiating cross-domain access, which is stored by the TPD module of the vehicle.

- 2) Upon receiving the message M_1 , OAS_{CA}^A will first check the freshness of the timestamp TS_1 based on the condition $|TS_{Now} - TS_1| < \Delta T$. If the condition is met, OAS_{CA}^A will compute the symmetric key $symk$ based on the home domain's authentication scheme. It will then decrypt and obtain $\langle Num, GID_j, ID_i^A, vpk_i^A \rangle$, and verify the validity of the vehicle V_i^A 's signature $Sign_{v_i}$. After successful verification, OAS_{CA}^A calculates a cross-domain identity $CID_i^A = H(ID_i^A, GID_B)$ (taking access to domain B as an example) for the vehicle. It selects a random number w_1 from Z_p^* , computes $W_1 = g^{w_1}$, $W^* = APK_B^{w_1}$, and a value E' encrypted with the symmetric key W^* which is $E' = E_{W^*}(vpk_i^A, GID_A, CID_i^A)$. Then, it calculates a hash value $Q' = H_1(vpk_i^A, GID_A, CID_i^A, TS_2)$, and computes a signature $F' = w_1 + ask_A \cdot Q' \pmod{q}$. Finally, the message $M_2 = \{W_1, E', F', TS_2\}$ is then forwarded to the OAS_{TA}^B of domain B. OAS_{CA}^A can aggregate and sign messages based on the destination domain identifier GID_j of the requested cross domain vehicle to improve authentication efficiency.

- 3) Upon receiving the message M_2 , OAS_{TA}^B checks whether $|TS_{Now} - TS_2| < \Delta T$. If the condition is met, it calculates the symmetric key $W^* = W_1^{ask_B} = (g^{ask_B})^{w_1}$. It further decrypts to obtain $\langle vpk_i^A, GID_A, CID_i^A, TS_2 \rangle$, and then checks if $g^{F'} = W_1 \cdot APK_A^{Q'}$ to determine the validity of signature F' . If the verification is successful, OAS_{TA}^B generates a polynomial $f_i(x, y)$ for vehicle V_i^A and selects a random element h from G_1 . Utilizing the domain's group key, it computes the license $Lics_i^A = (\sigma_1^A, \sigma_2^A) = (h, h^{x_B} (vpk_i^A)^{y_B})$, calculates the hash value $Q'' = H_2(Lics_i^A, f_i(x, y), TS_3)$, and finally chooses a random number w_2 to compute $W_2 = g^{w_2}$, $W^{**} = (vpk_i^A)^{w_2}$, and encrypts the value E'' using the symmetric key W^{**} which is $E'' = E_{W^{**}}(Lics_i^A, f_i(x, y))$. It then computes the signature $F'' = w_2 + ask_B \cdot Q'' \pmod{q}$, and sends the message $M_3 = \{W_2, E'', F'', TS_3\}$ to OAS_{CA}^A of domain A. Simultaneously, it uploads the vehicle access information to the blockchain by invoke Algorithm 1, i.e., $\{vpk_i^A, GID_A, GID_B, EXP\}$, where EXP represents the validity period of the license.
- 4) Upon receiving message M_3 , OAS_{TA}^A verifies the validity of timestamp TS_3 and checks if the data is recorded in the blockchain by invoke smart contract. If so, calculates W^{**}

Algorithm 1: Cross-Domain Vehicle Identity Registration.

Input: $vpk_i^A, GID_A, GID_B, EXP$
Output: bool

- 1: // Dom_Info is a struct used to store addresses and identifiers of all domains
- 2: **Struct** Dom_Info {
- 3: Address Daddr;
- 4: String GIDX;
- 5: }
- 6: **if** msg.sender not in Dom_Info.Daddr or GID_B not equal Dom_Info.GIDX **then**
- 7: **return** false
- 8: **end if**
- 9: // vehicles is a mapping type
- 10: // Determine if 'vehicles' contains vpk_i^A by determining that the field EXP is not empty
- 11: **if** vehicles[vpk_i^A]. $EXP > 0$ **then**
- 12: **return** false
- 13: **else**
- 14: vehicles[vpk_i^A] = [GID_A, GID_B, EXP]
- 15: **return** true
- 16: **end if**

$= W_2^{ask_A} = (g^{ask_A})^{w_2}$, and decrypts get $f_i(x, y)$, then calculates $Q'' = H(Lics_i^A, f_i(x, y), TS_3)$, and checks whether the equation $g^{F''} = W_2 \cdot APK_B^{Q''}$ holds true. If the verification equation holds, calculates the polynomial fragments $g_i(y) = f_i(ID_i^A, y)$, $W^{***} = (vpk_i^A)^{ask_A}$, and $E''' = E_{W^{***}}(Lics_i^A, g_i(y))$ for the vehicle, using the key $symk$ signs the message to obtain $Sign'_{v_A}$ and sends it to the vehicle V_i^A , that is, $M_4 = \{E''', Sign'_{v_A}, TS_4\}$.

5) Upon receiving the message M_4 , vehicle V_i^A verifies the validity of timestamp TS_4 and the validity of the signature $Sign'_{v_A}$, calculates $W^{***} = APK_A^{vsk_i^A}$, decrypts to get $\langle Lics_i^A, g_i(y) \rangle$ and stores it.

After completing the above steps, the authentication process for vehicle access to domain B is complete. It should be noted that the above process is for obtaining license when a vehicle needs to make its first cross-domain access. This process requires only a single round of interaction and can be pre-completed. When the vehicle needs to access foreign domain B again, it only needs to check whether the licenses have expired. If they haven't expired, the vehicle can directly engage in cross-domain communication. Otherwise, the vehicle sends a message to access domain B. Domain B's OAS_{TA}^B invokes the smart contract Algorithm 2 to query the validity of the vehicle's public key vpk_i^A . If the key is valid, the expiration date is updated, where $Nowtime$ indicates the current time, not the blockchain network time, new_EXP indicates the new expiration time. Additionally, if the group public key has been updated, the licenses must also be updated. The renewed licenses, denoted as $Lics'$, are then returned to the vehicle after encryption.

2) *Vehicle Cross-Domain Communication: Message signature:* After the vehicle completes the first-stage authentication and obtains a license $Lics_i^A$, let's assume it needs to send a

Algorithm 2: Vehicle Registration Information Update.

Input: $vpk_i^A, Nowtime, new_EXP$
Output: bool

- 1: **if** msg.sender not in Dom_Info.Daddr or GID_B not equal Dom_Info.GIDX **then**
- 2: **return** false
- 3: **end if**
- 4: // Determine if vpk_i^A exists and the expiration date EXP has expired
- 5: **if** vehicles[vpk_i^A]. $EXP > 0$ and vehicles[vpk_i^A]. $EXP \leq Nowtime$ **then**
- 6: vehicles[vpk_i^A]. $EXP = new_EXP$
- 7: **return** true
- 8: **else**
- 9: **return** false
- 10: **end if**

Algorithm 3: Message Signing.

Input: msg, σ_1, σ_2

- 1: choose random $\nu, \tau \in Z_p^*$
- 2: compute $I_0 = \sigma_1^\nu, I_1 = \sigma_2^\nu, I_2 = I_0^{vsk_i^A}, I_3 = g^\tau, I_4 = vpk_i^A \cdot APK_j^\tau$
- 3: choose random $t_k, t_v \in Z_p^*$
- 4: compute auxiliary values $R_0 = I_0^{t_v}, R_1 = g^{t_k}, R_2 = g^{t_v} \cdot APK_j^{t_k}$
- 5: challenge value $ch = H(I_0 \parallel I_1 \parallel I_2 \parallel I_3 \parallel I_4 \parallel R_0 \parallel R_1 \parallel R_2 \parallel msg)$
- 6: compute $z_k = (t_k + \tau \cdot ch) \bmod q$, $z_v = (t_v + vsk_i^A \cdot ch) \bmod q$

Output: $\{msg, \gamma = (I_0, I_1, I_2, I_3, I_4, ch, z_k, z_v)\}$

message to a vehicle V_i^B within access domain B. To prevent malicious behavior by the vehicle or potential attacks on the regular communication channel, the vehicle needs to perform the signature steps by Algorithm 3 to ensure the secure transmission of the message msg .

First, in steps 1–2 of Algorithm 3, the vehicle selects two random numbers $\nu, \tau \in Z_p^*$, then computes five randomized values I_0, I_1, I_2, I_3 and I_4 . Second, in steps 3–4, randomly select two numbers $t_k, t_v \in Z_p^*$, and compute auxiliary R_0, R_1 and R_2 . Third, in steps 5–6, calculate a challenge value ch and generate two signature values z_k and z_v . Finally, algorithm output message $M_v = \{msg, \gamma = (I_0, I_1, I_2, I_3, I_4, ch, z_k, z_v)\}$.

Message verification: When vehicle V_i^B or other RSUs receives the signature message M_v , it can independently verify the message by Algorithm 4. If algorithm output "1", it means the message verification passed and accepted. Otherwise, it fails and the message is rejected.

3) *Identity Tracking:* While safeguarding vehicle privacy information, it is also essential to ensure that in case of malicious actions by vehicles, such as sending false messages to occupy road resources, the cross-domain group administrator, namely OAS_{TA}^B , can trace the identity information of the vehicles. Specifically, upon receiving the message M_v , using the master

Algorithm 4: Message Verification.

Input: msg, γ , X_B , Y_B , APK_B

- 1: **if** $e(I_1, \hat{g}) = e(I_0, X_B) \cdot e(I_2, Y_B)$ **then**
- 2: compute $R'_0 = I_0^{z_v} I_2^{-ch}$, $R'_1 = g^{z_k} I_3^{-ch}$, $R'_2 = (g^{z_v} \cdot APK_B^{z_k}) I_4^{-ch}$
- 3: **if** $H(msg \parallel I_0 \parallel I_1 \parallel I_2 \parallel I_3 \parallel I_4 \parallel R'_0 \parallel R'_1 \parallel R'_2) = ch$ **then**
- 4: Accept msg
- 5: **else**
- 6: Reject msg
- 7: **end if**
- 8: **else**
- 9: Reject
- 10: **end if**

Output: '0' or '1'

Algorithm 5: Vehicle Identity Insert Illegal List.

Input: ID_i^A

Output: bool

- 1: // Raddr is an address array
- 2: **if** msg.sender not in Raddr **then**
- 3: **return** false
- 4: **end if**
- 5: // illegal_list is an uint array
- 6: illegal_list.push(ID_i^A)
- 7: **return** true

Algorithm 6: Vehicle Register Information Deletion.

Input: vpk_i^A

Output: bool

- 1: **if** msg.sender not in Dom_Info.Daddr or GIDB not equal Dom_Info.GIDX **then**
- 2: **return** false
- 3: **end if**
- 4: // Determine if vpk_i^A exists
- 5: **if** vehicles[vpk_i^A].EXP > 0 **then**
- 6: delete vehicles[vpk_i^A]
- 7: **return** true
- 8: **end if**

key ask_B to compute $vpk_i^A = I_4/I_3^{ask_B}$. The local database is then queried to obtain the identity CID_i^A . Further evidence is submitted to OAS_A^A in domain A to acquire the unique identity ID_i^A of vehicle V_i^A . Smart contract Algorithm 5 is invoked to add this identity to the illegal list within the consortium blockchain consisting of RSUs in the domain. Through Algorithm 6, the vehicle's cross-domain public key vpk_i^A is removed from the DS blockchain. As a result, our scheme provides non-repudiation for signed messages and the ability to trace vehicle identities.

4) *Identity Revocation:* To completely revoke malicious vehicles and prevent them from continuing to receive services

within the access domain, the group public key must be updated to update the license of legitimate vehicle. It also need be updated regularly to ensure the communication security among the group members. OAS_{TA}^B first selects a secret value d from Z_p^* , and computes the new group public key: $X_B' = X_B^d$, $Y_B' = Y_B^d$, and $mt = H(X_B', Y_B', TS)$, where TS represents the timestamp. Then two univariate polynomials are computed: $R(a) = (a - ID_1)(a - ID_2) \dots (a - ID_i)$ and $S(a) = f(a, mt) + R(a)d$, where ID_i is the identity of the vehicle in the illegal list. Finally, using the master private key u_j , the signature $F = \mu + u_j H(mt, S(a))$ is computed, where $\mu \in Z_p^*$. The composed message $\{X_B', Y_B', S(a), F, TS\}$ is broadcasted, and $S(a)$ is written into the blockchain. This prevents situations where the broadcasted message is not received by all legitimate member vehicles within the group. Vehicles can independently query the RSU blockchain to obtain the necessary information.

Upon receiving the message, the vehicle first verifies the validity of the timestamp TS and the signature F . If the verification succeeds, the vehicle queries the blockchain to retrieve all identities ID_i in the illegal list. It calculates $R(a)$ and polynomial fragment $g_i(mt)$ accordingly. The vehicle's OBU then sends $\{R(ID_i), g_i(mt)\}$ to the TPD module, which calculates $d = \frac{S(ID_i) - g_i(mt)}{R(ID_i)}$. This can prevent malicious vehicles from leaking secret values d . Lastly, the licenses are updated to $\{\sigma_1, \sigma_2^d\}$.

As $R(ID_i)$ is computed for vehicles in the illegal list is always 0, even if the value of the polynomial fragment $g_i(mt)$ is known, it's impossible to calculate the secret value d . This ensures that only legitimate vehicles can update their licenses, then revoke the malicious vehicle. Additionally, when the group key is periodically updated without needing to revoke vehicles, the operational process remains the same as described above.

V. SECURITY PROOF AND ANALYSIS

In this section, we first provide a security proof to demonstrate the EU-CMA (Existential Unforgeability against chosen-message attacks) security of the proposed scheme. Following that, we conduct a security analysis to establish the capability of VANETs cross-domain authentication scheme to resist various attacks.

A. Security Proof

Theorem 1: If the CDH assumption can not be solved, then there does not exist a probabilistic polynomial-time adversary $\mathcal{A}_I$ capable of breaking the EU-CMA security of our scheme.

Proof: If there exists adversary $\mathcal{A}_I$ capable of breaking the proposed scheme with a non-negligible probability ϵ_I , then it is possible to construct an algorithm $\mathcal{B}_I$ to utilize $\mathcal{A}_I$ for solving the CDH assumption. Given a group G and a CDH instance $\{\mathcal{X} = g^x, \mathcal{Y} = g^y, \mathcal{Z} = g^{ab}, \text{ where } x, y, a, b \in Z_p^*\}$, $\mathcal{B}_I$ simulates $\mathcal{A}_I$'s query oracle as follows:

Setup: Upon receiving the setup query from $\mathcal{A}_I$, $\mathcal{B}_I$ sends public parameter set $\{G_1, G_2, g, \hat{g}, H_i (i = 1, 2, 3)\}$ to $\mathcal{A}_I$.

H₃-query: For the query, $\mathcal{B}_I$ produces an empty list L_{H_3} . After receiving the query with the $\langle I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, msg \rangle$ from $\mathcal{A}_I$, $\mathcal{B}_I$ checks whether the set $\langle I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, msg \rangle$ is in the list L_{H_3} or not. If not, $\mathcal{B}_I$ chooses a random number $\tau_{H_3} \in Z_p^*$, adds the set $\langle I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, msg, \tau_{H_3} \rangle$ into the list L_{H_3} and returns $\tau_{H_3} = H_3(I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, msg)$ to $\mathcal{A}_I$. Otherwise, $\mathcal{B}_I$ returns τ_{H_3} to $\mathcal{A}_I$.

Extract license query: For the query, $\mathcal{B}_I$ produces an empty list L_{lics} . After receiving the query with the $\langle vpk_i^A, CID_i^A \rangle$ from $\mathcal{A}_I$, $\mathcal{B}_I$ checks whether the set $\langle vpk_i^A, CID_i^A \rangle$ is in the list L_{lics} or not. If not, $\mathcal{B}_I$ chooses two elements $\tau_{lics} = \tau_{\sigma_1^A}, \tau_{\sigma_2^A} \in G_1$ adds the set $\langle vpk_i^A, CID_i^A, \tau_{\sigma_1^A}, \tau_{\sigma_2^A} \rangle$ into the list L_{H_3} and returns $H_3 = (\sigma_1^A)^{x_B} (vpk_i^A)^{y_B}$ to $\mathcal{A}_I$. Otherwise, $\mathcal{B}_I$ returns τ_{lics} to $\mathcal{A}_I$.

Sign-query: After receiving the query with $M_i = \{msg, lics_i^A, vpk_i^A\}$ from $\mathcal{A}_I$ (assume that $\mathcal{A}_I$ has already executed extract license query with $\langle lics_i^A \rangle$), $\mathcal{B}_I$ first queries the set L_{lics} . If there have $lics_i^A$, $\mathcal{B}_I$ aborts the process. Otherwise, $\mathcal{B}_I$ chooses four random numbers $\nu, \tau, t_k, t_v \in Z_p^*$, and computes signature message $\langle I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2 \rangle$. If $ch = H_3(I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, msg)$ has already been existed in L_{H_3} , then $\mathcal{B}_I$ halts. Otherwise, $\mathcal{B}_I$ adds the set $\langle I_0, I_1, I_2, I_3, I_4, R_0, R_1, R_2, ch \rangle$ into the list L_{H_3} . And returns $\langle I_0, I_1, I_2, I_3, I_4, ch, z_k, z_v \rangle$ on the M_i to $\mathcal{A}_I$.

Forger: After sending the above queries, $\mathcal{A}_I$ outputs a forged signature $\langle I_0 = \sigma_1^\nu, I_1 = \sigma_2^\nu, I_2 = I_0^{vsk_i^A}, I_3 = \mathcal{X}, I_4 = vpk_i^A \cdot \mathcal{Z} \rangle$ on the $\langle msg, lics_i^A, vpk_i^A \rangle$. If $I_4 \neq vpk_i^A \cdot g^{xy}$, $\mathcal{A}_I$ aborts, $\mathcal{B}_I$ did not solve CDH. Otherwise, it meets the verification equation. $\mathcal{B}_I$ outputs the valid solution of CDH.

Assume that ϵ' denotes $\mathcal{B}_I$ did not abort during query-
ing phase, then $\Pr[\epsilon'] = (1 - \tau)^{q_s + q_{H_3} + q_{lic}}$, where q_s denotes the maximum times of sign queries, q_{H_3} denotes the maximum times of H_3 -query, q_{lic} denotes the maximum times of extract license queries. Assume that ϵ'' denotes $\mathcal{B}_I$ did not abort in the forgery phase, then $\Pr[\epsilon''] = \tau$. Hence, the probability that $\mathcal{B}_I$ does not abort in the simulation process is $\Pr[\epsilon' \wedge \epsilon''] = (1 - \tau)^{q_s + q_{H_3} + q_{lic}}$, where $\tau = \frac{1}{q_s + q_{H_3} + q_{lic}}$. When $q_s + q_{H_3} + q_{lic}$ is large enough, $(1 - \frac{1}{q_s + q_{H_3} + q_{lic} + 1})^{q_s + q_{H_3} + q_{lic} + 1}$ tends to e^{-1} . The probability that $\mathcal{B}_I$ does not abort during the simulation is at least $(1 - \tau)^{q_s + q_{H_3} + q_{lic}}$. Therefore, adversary $\mathcal{A}_I$ forges the signature scheme with a non-negligible probability ϵ_I , and simulator $\mathcal{B}_I$ outputs the valid solution of CDH with a non-negligible probability of no less than $\frac{\epsilon_I}{e(q_s + q_{H_3} + q_{lic})}$.

Theorem 2: If the LRSW Assumption can not be solved, then there does not exist a probabilistic polynomial-time adversary $\mathcal{A}_{II}$ capable of breaking the cross-domain license generation process of our scheme.

Proof: If there exists adversary $\mathcal{A}_{II}$ capable of breaking the license generation process of the proposed scheme with a non-negligible probability $\epsilon_{II} \geq 6(q + q_G)^2/p$, i.e., generate a valid license $Lic_i^A = (h, h^{x_j}(vpk_i^{A^{y_j}}))$, then it is possible to construct an algorithm $\mathcal{B}_{II}$ to utilize $\mathcal{A}_{II}$ for solving the LRSW Assumption [39].

B. Security Analysis

This subsection provides a detailed analysis of various security properties satisfied by the proposed scheme in a non-mathematical way.

- 1) *Mutual authentication:* In the vehicle cross-domain pre-authentication process, only one round of interaction between the vehicle and the OAS can be performed to authenticate each other. This is because messages are sent after being computed using encryption or signature algorithms, and the same applies to V2V and V2I communication processes. Additionally, under the CDH assumption and LRSW assumption, there is no adversary capable of forging messages sent by the vehicle and OAS. As a result, our scheme can achieve mutual authentication, ensuring the authenticity and integrity of messages.
- 2) *Conditional privacy-preserving:* As vehicles adopt an existing authentication scheme within home domain during the cross-domain pre-authentication phase, we will not discuss this part. Instead, the focus is on how vehicles can securely interact with other vehicles or roadside infrastructure in the foreign domain after crossing over. In the authentication phase, vehicles have already obtained licenses from the access domain. To preserve privacy, the vehicle can self-generate a one-time anonymous message γ according to these licenses. These messages do not contain any information related to the vehicle's true identity, ensuring the anonymity of the messages. In addition, as ν, τ, t_k and t_v are four random numbers and vsk_i^A is unknown to the attacker, the public key vpk_i^A of the vehicle cross-domain communication cannot be extracted from the one-time anonymous message, nor can it be linked to multiple one-time messages, ensuring unlinkability of messages. Simultaneously, in order to protect the interests of the access domain, when malicious vehicles are detected, the access domain OAS_{TA}^B can trace back to the vehicle's public key vpk_i^A , thus obtaining its real identity ID_i^A and verifying whether $CID_i^A = H(ID_i^A, GID_B)$ is correct. In addition, for vehicles in the illegal list, the illegal vehicles are revoked by updating the group public key to ensure the security of the system. Therefore, our scheme achieves identity anonymity, unlinkability, traceability and revocability for cross-domain vehicles. This guarantees vehicle privacy while also guarantees the legitimate rights of the access domain.
- 3) *Resistance to man-in-the-middle attack:* Assuming attacker $\mathcal{A}$ is able to intercept transmitted messages M_1, M_2, M_3, M_4 , and M_v during the cross-domain vehicle communication process. The attacker attempts to modify the intercepted messages to deceive OAS_{CA}^A, OAS_{TA}^B , and V_i^B . For instance, in the case of message M_2 , the attacker would need to acquire the secret values w_1, ask_A, CID_i^A , and vpk_i^A in order to generate a legitimate request message. The same applies to other transmitted messages, as any modifications to the message content would fail to pass the checks conducted by the recipients. Consequently, the proposed scheme is capable of resisting man-in-the-middle attacks.

TABLE II
COMPARISON OF FUNCTIONALITY FEATURES

Functionality	[19]	[18]	[23]	Our
Different configuration domains	×	×	×	✓
Integrity and authentication	✓	✓	✓	✓
Anonymity	✓	✓	×	✓
Unlinkability	✓	×	×	✓
Traceability	✓	✓	✓	✓
Identity revocation	×	✓	✓	✓
Man-in-the-middle attack	✓	✓	✓	✓
DDoS attack	✓	✓	×	✓
Collusive attack	×	×	✓	✓
Replay attack	✓	✓	✓	✓
Impersonation attack	✓	✓	✓	✓

- 4) *Resistance to DDos attack*: Due to the proposed scheme utilizing blockchain (such as Ethereum and Hyperledger) for cross-domain identity management, which eliminates the reliance on third-party organizations, it provides sufficient robustness to withstand DDos attacks. Therefore, the scheme is resistant to DDos attacks.
- 5) *Resistant to collusive attack*: Due to the different communication public keys and identities of vehicles when accessing different domains in the scheme. This avoids the collusion of OAS-tagged vehicle access records to expose the privacy of the vehicle. In addition, for DSs or RSU, attackers need to collude with more than half of the server nodes in the blockchain to tamper with relevant records and deceive the accessed domain. Therefore, our scheme can resist collusion attacks.
- 6) *Resistance to replay attack*: Our scheme adds timestamps to the messages being sent. The recipient will check the freshness of the message through an equation based on $|TS_{Now} - TS| < \Delta T$. If the equation is not satisfied, the message cannot be verified, then resisting replay attacks.
- 7) *Resistance to impersonation attack*: In our scheme, the attacker has the following two attack methods:
- Case 1*: OAS impersonation attack: Assume that the attacker captures the valid message M_2 sent by OAS_{CA}^A . As the message contains w_1 , ask_A , CID_i^A , and vpk_i^A , which are unknown to the attacker, and under the CDH assumption, the attacker cannot compute an effective request message M_2 . The same applies to other messages in the authentication phase.
- Case 2*: Vehicle impersonation attack: Assume that the attacker captures the valid message M_v sent by the vehicle. The attacker can forge a valid cross-domain license Lic_i^A and effective signature γ , which contradicts the LRSW assumption and CDH assumption. That is, the attacker cannot effectively message M_v within polynomial time. Therefore, our scheme can resist impersonation attacks.

C. Comparison of Security and Functionality

Table II presents a comparison of the functional characteristics of the related schemes [19], [18], [23] with the proposed scheme. The symbol "✓" indicates that the scheme is resistant to the attack or has the particular functionality, while the symbol "×" indicates the presence of an attack or the absence of the functionality. According to the Table II, only our scheme is

capable of providing more functions and has better security characteristics compared to the existing schemes [19], [18] and [23].

VI. PERFORMANCE EVALUATION

In this section, we use blockchain prototypes to validate the feasibility of our scheme in an Ethereum test network. The environment consists of two administrative domains, each consisting of an OAS, some RSU and vehicle OBU entities. Additionally, there is also a server cluster made up of three DSs entities. Based on the simulation results, we analyzed the performance of the proposed scheme from three aspects: communication cost, computing cost, and blockchain gas cost.

A. Experiment Configuration and Implementation

1) *Environment Configuration*: The simulation experiment runs on a 64-bit Ubuntu.04 host server equipped with an i7 11700X CPU @ 2.50 GHz and 16 GB of memory. The entity operations of the OAS, DS and terminal run directly on the server. To evaluate the performance of the scheme, we implemented it using C++. We applied the Miracl encryption library [33] with BN curves to provide a 128-bit security level. Additionally, to implement the blockchain-supported prototype, we first used the online public Ethereum test network (Georli) to deploy Solidity smart contracts. We connected Remix and Georli using the Google Chrome extension MetaMask-Chrome and deployed the compiled smart contracts to Remix to measure the gas costs of each algorithm. Gas costs reflect the cost of using Ethereum-type blockchain [34].

2) *Comparison Scheme Configuration*: To demonstrate the effectiveness of the proposed scheme, we compared it with three other schemes. In all schemes, vehicles in different domains for VANETs must be authenticated through an edge server before communication.

- In Li et al.'s scheme [19], V_i^A directly verifies the identity through edge server ES_B and chain queries, and negotiates a session key with V_i^B . For fair cost comparison, only one round of signature message from V_i^A to V_i^B is included in the session key phase.
- In Yang et al.'s scheme [18], V_i^A generates necessary cross-domain information with the help of edge server ES_A and blockchain, and signs the message msg . ES_B assists in verifying the identity through chain queries, and V_i^B verifies the identity of the received message through ES_B .
- In Yang et al.'s scheme [23], V_i^A is authenticated through edge server ES_X (multiple edge servers) and generates necessary cross-domain information. ES_B authenticates the V_i^A across domains. The scheme only includes the vehicle cross-domain authentication process and does not involve the V2V and V2I communication processes.

B. Gas Cost

To verify the correctness of our designed smart contract algorithm and analyze the cost of gas in blockchain, we conducted testing based on Ethereum, and all functions were successfully

TABLE III
THE COSTS OF ALL SMART CONTRACT FUNCTIONS

Function	Gas	Ether
deploy contract	1275353	0.00473706
register	94351	0.00030878
query	24966	0.00008028
update	30756	0.000093
insert	70540	0.00022025
delete	31076	0.00008831

tested. On June 20, 2023, the average gas price of Ethereum was 19.2 Gwei, and the cost of all operations is shown in Table III. The largest cost is the deployment of smart contracts, but they are only executed once.

C. Communication Overhead

According to the experimental setup, for group G_1 in bilinear mapping, the elements size in G_1 is 64 bytes, $|Z_p^*|$ is 32 bytes, the size of timestamp is set to 4 bytes, ordinary hash function and symmetric encryption/decryption are 32 bytes and 20 bytes, respectively. Assuming that the length of the vehicle signature message in the original domain is 20 bytes. We mainly consider the message length in the two stages of cross domain pre-authentication and communication for VANETs. For the sake of convenience, we do not calculate the length of the sent message msg . In addition, let I_{v-s} , I_{s-s} , I_{s-bc} , and I_{v-v} represent the interactions between vehicles and servers (RSU/ES/OAS), servers and servers, servers and blockchain, and vehicles and vehicles, respectively.

The vehicle cross-domain pre-authentication phase (CDPAP) in our scheme can be completed in one round of interaction, including messages $M_1 = 20 + 20 + 4 = 44$ bytes, $M_2 = 64 + 20 + 32 + 4 = 120$ bytes, $M_3 = 64 + 20 + 32 + 4 = 120$ bytes, $M_4 = 20 + 20 + 4 = 44$ bytes, and the costs for writing to the blockchain: $64 + 4 + 4 + 4 + 4 = 80$ bytes, query costs: $64 + 20 + 4 = 88$ bytes, and the total communication cost: $2I_{v-s} + 2I_{s-s} + 2I_{s-bc} = 496$ bytes. In the vehicle cross-domain communication phase (CDCP), the communication cost is: $I_{v-v} = 5 * 64 + 3 * 32 = 416$ bytes, and the total communication cost is: $2I_{v-s} + 2I_{s-s} + 3I_{s-bc} + I_{v-v} = 912$ bytes. For scheme [19]: the cost of CDPAP is: $2I_{v-s} + 2I_{s-bc} = 704$ bytes, the cost of CDCP is: $I_{v-v} = 924$ bytes. The total cost is $2I_{v-s} + 2I_{s-bc} + I_{v-v} = 1628$ bytes. For scheme [18]: the cost of CDPAP is: $2I_{v-s} + I_{s-s} + 2I_{s-bc} = 576$ bytes, the cost of CDCP is: $I_{v-v} + 2I_{v-s} + 2I_{s-bc} = 644$ bytes. The total cost is $3I_{v-s} + I_{s-s} + 4I_{s-bc} = 1220$ bytes. For scheme [23]: the cost of CDPAP is: $3I_{v-s} + t * I_{s-s} = 436 + 140t$ bytes (t represents the number of servers). The comparison of the communication cost in CDPAP and CDCP is shown in Fig. 4, where t is set to 1.

Based on the above analysis and Fig. 4, Table IV, it can be seen that the proposed scheme has lower communication costs in the two stages of vehicle cross-domain pre-authentication and cross-domain communication compared to other solutions. Specifically, compared with [19] and [18], the reduction on total communication overhead are about $(1628 - 912)/1628 \approx 43.98\%$, $(1220 - 912)/1220 \approx 25.25\%$, respectively. Compared

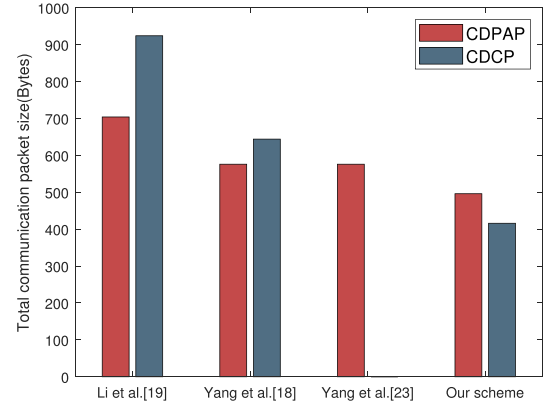


Fig. 4. Comparison of related communication packet size.

TABLE IV
COMPARISON OF COMMUNICATION PACKET SIZE (BYTES)

Schemes	CDPAP	CDCP	Total
Li et al.[19]	704	924	1628
Yang et al.[18]	576	644	1220
Our scheme	496	416	912

with [23], the reduction on CDPAP communication cost is about $(576 - 496)/576 \approx 13.89\%$.

D. Computational Overhead

To compare the computational cost of the proposed scheme with several representative cross domain schemes in VANETs environment, we defined symbols for some time-consuming operations, as shown in Table V. Note that we omitted some negligible computational time such as hash functions and the addition of large numbers.

During the cross-domain pre-certification process of the vehicle. In scheme [19], the vehicle needs to perform 10 modular exponentiation operations and 5 modular multiplication operations, so the computational cost is $10T_{me} + 5T_{mm} = 6.995$ ms. The server (RSU) needs to perform 5 modular exponentiation operations and 4 modular multiplication operations, so the computational cost is $5T_{me} + 4T_{mm} = 3.553$ ms. Therefore, in the vehicle cross-domain authentication stage, the total computational cost is $15T_{me} + 9T_{mm} = 10.548$ ms. In scheme [18], the vehicle needs to perform 3 elliptic curve scalar multiplication operations, the cost is $3T_{ecm} = 1.149$ ms, and the server performs 4 elliptic curve scalar multiplication operations and one point addition operation, the cost is $4T_{ecm} + T_{eca} = 1.536$ ms, so the total computational cost is $7T_{ecm} + T_{eca} = 2.685$ ms. In scheme [23], the vehicle needs to perform at least 3 bilinear pairing operations and 3 bilinear scalar multiplication operations, the cost is $3T_{bp} + 3T_{bpm} = 7.167$ ms. The server side needs to perform 4 bilinear pairing operations and 3 bilinear pairing scalar multiplication operations, the cost is $4T_{bp} + 3T_{bpm} = 8.867$ ms, resulting in a total computational cost of $7T_{bp} + 6T_{bpm} = 16.034$ ms. In our scheme, the vehicle needs to perform two bilinear pairing scalar multiplication operations and two symmetric encryption/decryption operations, the cost is $2T_{bpm}$

TABLE V
THE EXECUTION TIME OF CRYPTOGRAPHIC OPERATIONS

Symbol	Description	Format	Time(ms)
T_{bp}	Bilinear pairing operation	$e(S, T)$, where $S, T \in G_1$	1.7
T_{bpm}	Scale multiplication operation related to the bilinear pairing	$g^x, g \in G_1, x \in \mathbb{Z}_p^*$	0.689
T_{aes}	Symmetric encryption/decryption operation(AES-CBC)	$E(k, \cdot)/D(k, \cdot)$	0.001
T_{pm}	Point multiplication operation	$P_1 * P_2$, where $P_1, P_2 \in G_1$	0.003
T_{btp}	Bivariate t-polynomial operation($t=100$)	$f(x, y) = \sum_{i,j=0}^t s_{ij}x^i y^j$	0.06
T_{ecm}	Scale multiplication operation related to the ECC	$x \cdot P$, where $P \in G$ and $x \in \mathbb{Z}_p^*$	0.383
T_{eca}	Point addition operation related to the ECC	$S' + T'$, where $S', T' \in G$	0.004
T_{mm}	A modular multiplication operation	$X * Y, X, Y \in \mathbb{Z}_{n^2}^*$	0.037
T_{me}	a modular exponentiation	$g^{r'} \bmod(n^2), g = (n+1) \in \mathbb{Z}_{n^2}^*, r \in \mathbb{Z}_n^*$	0.681

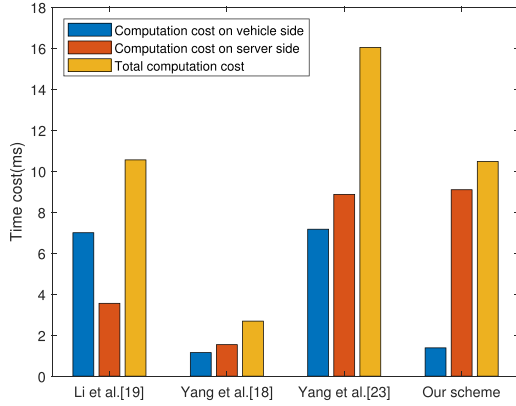


Fig. 5. Comparison of computation time cost in CDPAP.

$+ 2T_{aes} = 1.38$ ms, and the server needs to perform 13 bilinear pairing scalar multiplication operations, 6 Symmetric encryption/decryption operations, 3 point multiplication operations and two polynomial operations, the cost is $13T_{bpm} + 6T_{aes} + 3T_{pm} + 2T_{btp} = 9.092$ ms. The total computational cost is $15T_{bpm} + 8T_{aes} + 3T_{pm} + 2T_{btp} = 10.472$ ms. The comparison of the computation cost in vehicle side and server side is shown in Fig. 5.

During the cross-domain communication part of the vehicle. In scheme [19], vehicle sender needs to perform 3 modular exponentiation operations, one modular multiplication operations and one symmetric encryption/decryption operation, the cost is $3T_{me} + T_{mm} + T_{aes} = 2.081$ ms. Receiver needs to perform 5 modular exponentiation operations, 2 modular multiplication operations and one symmetric encryption/decryption operation, the cost is $5T_{me} + 2T_{mm} + T_{aes} = 3.48$ ms. The total computational cost is $8T_{me} + 3T_{mm} + 2T_{aes} = 5.561$ ms. In scheme [18], the cost of vehicle sender signatures can be neglected. Receiver verification needs to perform 3 elliptic curve scalar multiplication operations and 2 point addition operation, the cost is $3T_{ecm} + 2T_{eca} = 1.157$ ms. In our scheme, vehicle sender needs to perform 9 bilinear scalar multiplication operations and two point multiplication operations, the cost is $9T_{bpm} + 2T_{pm} = 6.207$ ms. Receiver verification needs to perform 3 bilinear pairing operations and 7 bilinear pairing scalar multiplication operations, the cost is $3T_{bp} + 7T_{bpm} = 9.923$ ms. The total computational cost is $15T_{bpm} + 2T_{pm} + 3T_{bp} = 16.13$ ms.

Based on the above analysis and Fig. 5, it can be seen that the proposed scheme has lower computational costs compared to schemes [19] [23] during VANETs cross-domain authentication stage, but higher than scheme [18]. However, the computational cost at the vehicle side is the lowest, only 1.38 ms. Although the computational overhead in the cross-domain communication stage is higher than other schemes, but our scheme is more comprehensive in functionality than other solutions, with the lowest communication overhead and acceptable computational costs. Therefore, the proposed scheme is generally superior to other solutions and more suitable for applications involving the VANETs.

VII. CONCLUSION

To address the security and privacy problems in cross-domain authentication for VANETs, we proposed a revocable cross-domain authentication scheme based on blockchain to protect vehicle communication. Our scheme allows secure cross-domain authentication between domains using different management technologies, which can be completed in only one round of interaction. Cross-domain identities are distributed and managed by smart contracts deployed on a blockchain system, thereby reducing the likelihood of a single point of failure. Additionally, in the V2V and V2I communication processes, the short random signature method is used to protect the identity privacy of the cross-domain vehicle, and the legitimate rights and interests of the accessed domain are guaranteed by revoking the identity of malicious vehicles. This scheme avoids real-time on-chain queries to meet the latency-sensitive requirements of VANETs. Security and performance analyses show that the proposed scheme meets the basic security and privacy requirements of VANETs and has certain advantages over the most advanced cross-domain authentication schemes for VANETs. In the future, we will further optimize the computing overhead of cross-domain communication and improve the overall performance in cross-domain scenarios.

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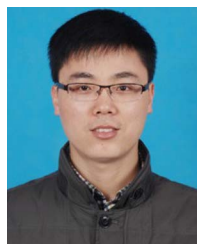
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REFERENCES

- [1] T. Alladi, V. Chamola, N. Sahu, V. Venkatesh, A. Goyal, and M. Guizani, "A comprehensive survey on the applications of blockchain for securing vehicular networks," *IEEE Commun. Surv. Tut.*, vol. 24, no. 2, pp. 1212–1239, Secondquarter 2022.
- [2] S. Valtolina, L. Ferrari, and M. Mesiti, "Ontology-based consistent specification of sensor data acquisition plans in cross-domain iot platforms," *IEEE Access*, vol. 7, pp. 176141–176169, 2019.
- [3] A. Lei, H. Cruickshank, Y. Cao, P. Asuquo, C. P. A. Ogah, and Z. Sun, "Blockchain-based dynamic key management for heterogeneous intelligent transportation systems," *IEEE Internet Things J.*, vol. 4, no. 6, pp. 1832–1843, Dec. 2017.
- [4] J. Zhang, H. Zhong, J. Cui, Y. Xu, and L. Liu, "Smaka: Secure many-to-many authentication and key agreement scheme for vehicular networks," *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 1810–1824, 2021.
- [5] L. Wei, J. Cui, H. Zhong, I. Bolodurina, C. Gu, and D. He, "A decentralized authenticated key agreement scheme based on smart contract for securing vehicular ad-hoc networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4318–4333, May 2024.
- [6] J. Zhang, J. Cui, H. Zhong, Z. Chen, and L. Liu, "Pa-crt: Chinese remainder theorem based conditional privacy-preserving authentication scheme in vehicular ad-hoc networks," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 2, pp. 722–735, Mar./Apr. 2021.
- [7] V. Odelu, A. K. Das, and A. Goswami, "A secure biometrics-based multi-server authentication protocol using smart cards," *IEEE Trans. Inf. Forensics Secur.*, vol. 10, no. 9, pp. 1953–1966, Sep. 2015.
- [8] X. Zhang, H. Zhong, C. Fan, I. Bolodurina, and J. Cui, "Cbacs: A privacy-preserving and efficient cache-based access control scheme for software defined vehicular networks," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 1930–1945, 2022.
- [9] L. Wei, J. Cui, H. Zhong, I. Bolodurina, and L. Liu, "A lightweight and conditional privacy-preserving authenticated key agreement scheme with multi-ta model for fog-based vanets," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 1, pp. 422–436, Jan./Feb. 2023.
- [10] Y. Tan, J. Wang, J. Liu, and N. Kato, "Blockchain-assisted distributed and lightweight authentication service for industrial unmanned aerial vehicles," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 16928–16940, Sep. 2022.
- [11] C. Lai, H. Zhou, N. Cheng, and X. S. Shen, "Secure group communications in vehicular networks: A software-defined network-enabled architecture and solution," *IEEE Veh. Technol. Mag.*, vol. 12, no. 4, pp. 40–49, Dec. 2017.
- [12] X. Feng, Q. Shi, Q. Xie, and L. Wang, "P2ba: A privacy-preserving protocol with batch authentication against semi-trusted rsus in vehicular ad hoc networks," *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 3888–3899, 2021.
- [13] L. Wei, J. Cui, Y. Xu, J. Cheng, and H. Zhong, "Secure and lightweight conditional privacy-preserving authentication for securing traffic emergency messages in vanets," *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 1681–1695, 2021.
- [14] C. Lin, X. Huang, and D. He, "Ebcpa: Efficient blockchain-based conditional privacy-preserving authentication for vanets," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 3, pp. 1818–1832, May/Jun. 2023.
- [15] Y. Wang, X. Wang, H.-N. Dai, X. Zhang, and M. Imran, "A data reporting protocol with revocable anonymous authentication for edge-assisted intelligent transport systems," *IEEE Trans. Ind. Inform.*, vol. 19, no. 6, pp. 7835–7847, Jun. 2023.
- [16] J. Sun, G. Xu, T. Zhang, X. Cheng, X. Han, and M. Tang, "Secure data sharing with flexible cross-domain authorization in autonomous vehicle systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7527–7540, Jul. 2023.
- [17] Y. Lin, X. Wang, Q. Gan, and M. Yao, "A secure cross-domain authentication scheme with perfect forward security and complete anonymity in fog computing," *J. Inf. Secur. Appl.*, vol. 63, 2021, Art. no. 103022.
- [18] Y. Yang, L. Wei, J. Wu, C. Long, and B. Li, "A blockchain-based multidomain authentication scheme for conditional privacy preserving in vehicular ad-hoc network," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8078–8090, Jun. 2022.
- [19] X. Li, J. Liu, M. S. Obaidat, P. Vijayakumar, Q. Jiang, and R. Amin, "An unlinkable authenticated key agreement with collusion resistant for vanets," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 7992–8006, Aug. 2021.
- [20] R. Vilalta et al., "Vehicular message exchange in cross-border scenarios using public cloud infrastructure," in *Proc. IEEE 3rd 5G World Forum*, 2020, pp. 251–256.
- [21] C. Chen, F. Shi, H. Yu, and N. Fei, "Anonymous authentication based on cloud storage for cross-regional vehicles in vanet," in *Proc. 2016 IEEE Int. Conf. Ubiquitous Wireless Broadband*, 2016, pp. 1–4.
- [22] C. Xu, M. Ma, X. Huang, and H. Bao, "A cross-domain group authentication scheme for lte-a based vehicular network," in *Proc. IEEE 9th Int. Conf. Commun. Softw. Netw.*, 2017, pp. 595–599.
- [23] A. Yang, J. Weng, K. Yang, C. Huang, and X. Shen, "Delegating authentication to edge: A decentralized authentication architecture for vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 1284–1298, Feb. 2022.
- [24] L. Yang and J. Liu, "Cross domain authentication based on blockchain for mobile terminals in edge computing environment," in *Proc. 16th Int. Conf. Intell. Syst. Knowl. Eng.*, 2021, pp. 525–529.
- [25] M. Wang, L. Rui, Y. Yang, Z. Gao, and X. Chen, "A blockchain-based multi-ca cross-domain authentication scheme in decentralized autonomous network," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 3, pp. 2664–2676, Sep. 2022.
- [26] J. Chen, Z. Zhan, K. He, R. Du, D. Wang, and F. Liu, "Xauth: Efficient privacy-preserving cross-domain authentication," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 5, pp. 3301–3311, Sep./Oct. 2022.
- [27] L. Hua, H. Jiang, J. Xiao, and M. Samie, "Cross-domain self-authentication based consortium blockchain for autonomous valet parking system," *IEEE Access*, vol. 10, pp. 87950–87961, 2022.
- [28] Z. Xu, W. Liang, K.-C. Li, J. Xu, and H. Jin, "A blockchain-based roadside unit-assisted authentication and key agreement protocol for internet of vehicles," *J. Parallel Distrib. Comput.*, vol. 149, pp. 29–39, 2021.
- [29] Z. Ma, J. Zhang, Y. Guo, Y. Liu, X. Liu, and W. He, "An efficient decentralized key management mechanism for vanet with blockchain," *IEEE Trans. Veh. Technol.*, vol. 69, no. 6, pp. 5836–5849, Jun. 2020.
- [30] J. Sun, G. Xu, T. Zhang, M. Alazab, and R. H. Deng, "A practical fog-based privacy-preserving online car-hailing service system," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 2862–2877, 2022.
- [31] D. Pointcheval and O. Sanders, "Short randomizable signatures," in *Proc. Topics Cryptology - CT-RSA 2016*, K. Sako, ed. Cham, Springer International Publishing, 2016, pp. 111–126.
- [32] D. Dolev and A. Yao, "On the security of public key protocols," *IEEE Trans. Inf. Theory*, vol. 29, no. 2, pp. 198–208, Mar. 1983.
- [33] "Miracl core," [Online]. Available: <https://github.com/miracl/core>
- [34] G. W. DR., "Ethereum: A secure decentralised generalised transaction ledger," *Ethereum Project Yellow Paper*, vol. 151, no. 2014, pp. 1–32, 2014.



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